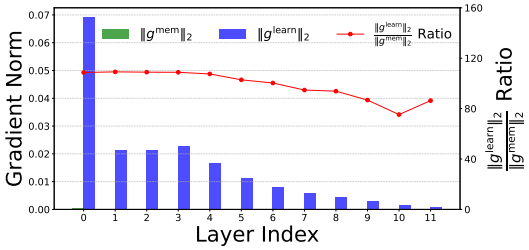
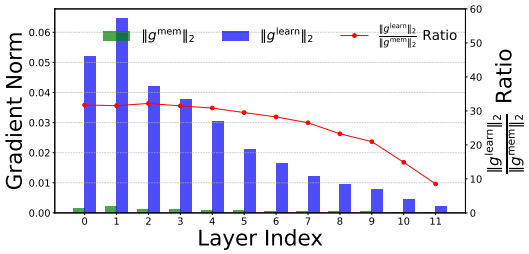
G.3.2 Vision Datasets Results

For the vision modality, it needs to be acknowledged that Post-LN architectures are not available in practice/literature, and only Pre-LN models are available. Hence, we provide additional experiments for Pre-LN models - ViT-B, ViT-S, and DeiT using multiple datasets - CIFAR10, CIFAR100, UTK-Face. The results for the same are presented in Figs. 31, 32, and 33.



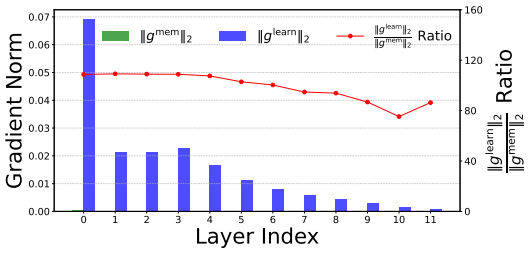
(a) Pre-LN (ViT-B) $\left\|g_x^{\text{learn}}\right\|_2$ & $\|g_x^{\text{mem}}\|_2$ analysis across layers

Figure 31: **Learning vs. Memorization Gradients in Pre-LN Models (CIFAR10 Dataset):** Results clearly exhibit high gradient norms of early layers LNs than later layers for both learning and memorization in Pre-LN (ViT-B) models. Importantly, the learning gradient norms ($\left\|g_x^{\text{learn}}\right\|_2$) are consistently higher than the memorization gradient norms ($\|g_x^{\text{mem}}\|_2$) across all layers.



(a) Pre-LN (ViT-S) $\left\|g_x^{\text{learn}}\right\|_2$ & $\|g_x^{\text{mem}}\|_2$ analysis across layers

Figure 32: **Learning vs. Memorization Gradients in Pre-LN Models (NICO++ Dataset):** Results clearly exhibit high gradient norms of early layers LNs than later layers for both learning and memorization in Pre-LN (ViT-S) models. Importantly, the learning gradient norms ($\left\|g_x^{\text{learn}}\right\|_2$) are consistently higher than the memorization gradient norms ($\|g_x^{\text{mem}}\|_2$) across all layers.



(a) Pre-LN (DeiT) $\left\|g_x^{\text{learn}}\right\|_2$ & $\|g_x^{\text{mem}}\|_2$ analysis across layers

Figure 33: **Learning vs. Memorization Gradients in Pre-LN Models (UTK-Face Dataset):** Results clearly exhibit high gradient norms of early layers LNs than later layers for both learning and memorization in Pre-LN (DeiT) models. Importantly, the learning gradient norms ($\left\|g_x^{\text{learn}}\right\|_2$) are consistently higher than the memorization gradient norms ($\|g_x^{\text{mem}}\|_2$) across all layers.

H Analysis across multiple Noisy Label Ratios and Optimizer

We extend our analysis to evaluate whether the claim—*removing LN parameters mitigates memorization in Post-LN models while impairing generalization in Pre-LN models*—holds consistently across higher noisy label ratios (2% and 5%) in the training dataset, as shown in Table 4.

Noise	Model	Setting	Learning (↑)	Memorization (↓)	Recovery (↑)	Random Prediction (↓)
2%	**Post-LN (BERT)**	Before	91.70	100.00	0.00	0.00
		After	92.00	20.62	76.25	3.12
	Pre-LN (GPT-Neo)	Before	91.35	100.00	0.00	0.00
		After	84.85	66.87	16.56	16.56
5%	**Post-LN (BERT)**	Before	90.35	100.00	0.00	0.00
		After	91.25	27.00	66.88	6.12
	Pre-LN (GPT-Neo)	Before	90.35	100.00	0.00	0.00
		After	82.60	67.50	11.00	21.50

Table 4: Results for higher noisy label ratios (2% and 5%) on the Emotions dataset.

Additionally, we assess the robustness of this finding under a different optimization setting by experimenting with Muon optimizer other than Adam, which was used in the primary experiments, as shown in Table 5.

Model	Setting	Learning (↑)	Memorization (↓)	Recovery (↑)	Random Prediction (↓)
Post-LN (BERT)	Before	91.95	100.00	0.00	0.00
	After	92.00	25.00	62.50	12.50
Pre-LN (GPT-Neo)	Before	91.70	100.00	0.00	0.00
	After	85.55	32.50	29.38	38.12

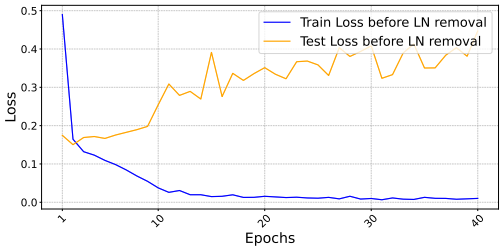
Table 5: Results with the Muon optimizer with Emotions Dataset.

Overall, the results demonstrate that our observations are agnostic to optimizer and noise-label ratios.

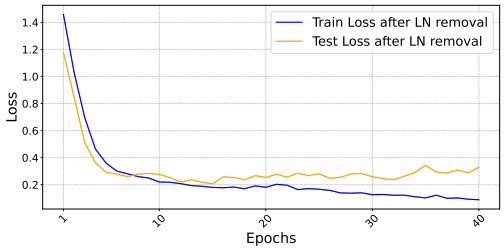
I Loss and Gradient Norms Analysis across Epochs

In Sec. 4, we showed how on LN parameters removal in Post-LN models, memorization is mitigated over epochs, while in Pre-LN models, the testing accuracy drops as training progresses, without recovering at any time. To further corroborate this claim, we plot the train-test loss functions for Post-LN and Pre-LN models, where the overfitting gap decreases in Post-LN models, while for Pre-LN models it increases, upon LN removal, as shown in Fig. 34.

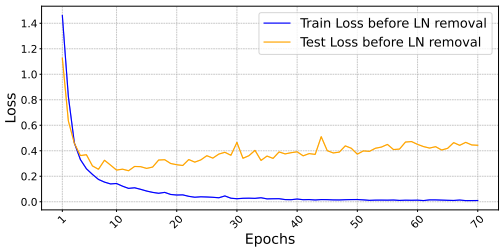
Apart from the loss curves, we also provide how the gradient norm evolves across epochs for both Post-LN and Pre-LN models in Fig. 35.



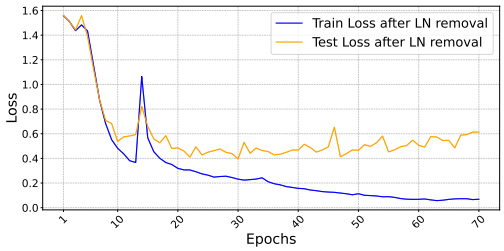
(a) BERT (Post-LN) before LN removal.



(b) BERT (Post-LN) after LN removal.

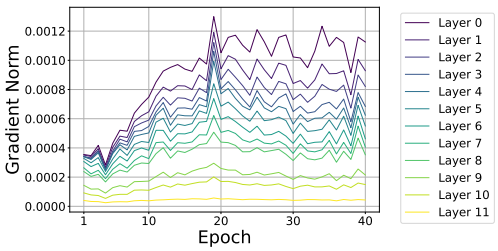


(c) GPT-Neo (Pre-LN) before LN removal.

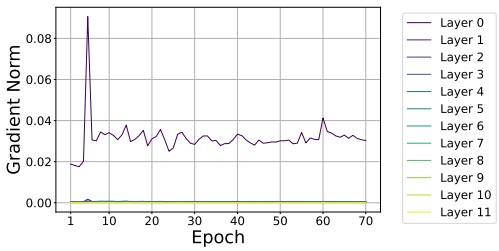


(d) GPT-Neo (Pre-LN) after LN removal.

Figure 34: Train and test loss trends across epochs for Post-LN (BERT) and Pre-LN (GPT-Neo) models before and after LayerNorm (LN) removal. In Post-LN models, LN removal reduces the overfitting gap (difference between train and test losses), mitigating memorization. In Pre-LN models, LN removal widens the overfitting gap, indicating impaired generalization.



(a) Gradient norms for BERT



(b) Gradient norms for GPT-Neo

Figure 35: Gradient norms gradually increase across epochs for both Pre-LN and Post-LN models, with early layers exhibiting the highest gradient norms.

J Validity of Results for Generative Modeling Task

To verify the robustness of our findings in a generative modeling context, we conduct a Next Token Prediction (NTP) experiment on the Emotions dataset using both BERT (Post-LN) and GPT-Neo (Pre-LN) models.

Each input sample is reformulated as:

$$\text{original text} + \text{“This emotion is [type]”}$$

where the model predicts the token corresponding to the emotion label $[type]$, one of six possible emotion types. To introduce **noisy labels**, we randomly replace the $[type]$ token for 1% of the training samples with a different emotion label. Two different model configurations are trained and evaluated: (i) before LN parameters removal, and (ii) after LN parameters removal.

As shown in Table 6, removing LN parameters substantially reduces memorization in Post-LN models (BERT), while impairing learning and generalization in Pre-LN models (GPT-Neo). This confirms that the trends reported in the main paper hold consistently for generative modeling tasks as well.